

Elevated skin site microbiome and dual-pathway hypothesis
for equine summer eczema

ELEVATED SKIN SITES (EtS) — normal microbial equilibrium

Deinococcaceae (Deinococcus spp.)

UV radiation & reactive oxygen species resistance

Carotenoid-mediated photoprotection [1]

Micrococcaceae (Micrococcus spp.)

Urea catabolism — balanced; skin pH ≈ 5

Acid mantle maintenance [2]

community
disruption

DYSBIOSIS TRIGGER

Culicoides spp. bite → IgE-mediated hypersensitivity → pro-inflammatory cytokines [3]

PATHWAY 1

Loss of Deinococcus

↓ UV photoprotection

↓ Reactive oxygen species scavenging

↓ Antioxidant defence capacity

→ Oxidative damage to keratinocytes

→ Impaired epidermal repair

→ Amplified local inflammatory response

[1]

PATHWAY 2

Micrococcus overgrowth / dysbiosis

↑ Urea hydrolysis → NH₃ + CO₂

↑ Ammonia accumulation in sweat film

↑ Skin surface pH (> 6.5)

→ Acid mantle disruption [2, 4]

→ Antimicrobial peptide inactivation

→ Compromised epidermal barrier

[2, 4]



SKIN BARRIER FAILURE

Chronic pruritus · Mast cell / IgE activation · Secondary microbial colonisation

Self-perpetuating itch-scratch cycle

EQUINE SUMMER ECZEMA (Sweet itch) [3]

Recurrent seasonal pruritic dermatitis; predilection for mane, tail base, and dorsum

Clinical signs: alopecia, excoriation, lichenification, secondary infection

[1] Battista JR (1997) Annu Rev Microbiol 51:203–224 — Deinococcus UV/ROS resistance
[2] Kloos WE & Musselwhite MS (1975) Appl Microbiol 31:381–385 — Micrococcus skin ecology
Fluhr JW & Darlenski R (2014) Curr Probl Dermatol 49:1–10 — urea, pH and skin barrier
[3] Schaffartzik A et al. (2012) Vet Immunol Immunopathol 147:113–126 — Culicoides hypersensitivity & sweet itch
[4] Visscher MO et al. (2000) Skin Pharmacol Appl Skin Physiol 13:140–149 — urea hydrolysis and skin pH